

TrackMind

GenAI Transparency Log

2758-T4 Advanced Topics in Machine Learning | Nova School of Business and Economics
Team L TXB | April 22 – May 13, 2026 | Masters Program in Business Analytics

Project	TrackMind – Railway TSI Compliance Intelligence Platform
Team Members	Filippo Deluchi (71064), Catarina Maria Anastacio Palma (56526), Afonso Jose Aviz De Brito Joao (72008), Paul Gereon Specht (72765), Nina Berenice Wecera (72603)
Course	2758-T4 Advanced Topics in Machine Learning, Nova SBE
Submission Date	May 13, 2026
AI Tools Used	Claude (Anthropic) — claude-sonnet-4-6 / claude-haiku-4-5 ChatGPT (OpenAI), Lovable, Codex / GPT-4.1 (OpenAI via VS Code)
Log Period	April 22, 2026 – May 13, 2026
Total Stages	37 documented AI-use stages (32 Claude / Filippo, Nina, Catarina, Afonso; 5 ChatGPT / Paul)

Overview

This log documents all usage of Claude (Anthropic) and ChatGPT (OpenAI) throughout the TrackMind project, from initial ideation in late April 2026 through technical implementation and pitch preparation in May 2026. It is submitted as the GenAI Transparency Appendix required under course 2758-T4 evaluation criteria.

The log covers two contributors and two tools. Nina used Claude throughout the project for ideation, research, technical build, and presentation work. Paul used ChatGPT for project alignment, backend architecture, demo reliability, and pitch deck refinement.

The log is structured chronologically by stage. Each entry records what was given to the AI tool, what the tool produced, and — critically — what human judgment, correction, or direction shaped or overrode that output. The final section compares AI contributions against the team's own inputs to give an honest picture of the collaboration.

All content produced with AI assistance was reviewed and edited by team members before use. No output was included in any deliverable verbatim without human review. Financial projections and market sizing figures were verified against primary sources including ERA publications and the ERADIS database.

Scope and Coverage

Nina's Claude usage covers 14 stages from April 22 through May 12, 2026. Filippo's Claude usage covers 1 stage from May 9 through May 12, 2026. Catarina's AI usage (Claude + Lovable + Codex/GPT-4.1) covers 5 stages in May 2026. Afonso's Claude usage covers 12 stages on May 13, 2026. All sessions materially affected the project deliverable.

Paul's ChatGPT usage covers 5 stages from May 3 through May 12, 2026. Excluded from Paul's log are interactions unrelated to TrackMind: generic VS Code setup, pure branch navigation, unrelated course projects (NOS prompt engineering, Ambulink, Lovable, DCA), personal chats, and minor

one-off wording edits. Only sessions that materially shaped the TrackMind backend, demo, or pitch deck are included.

Stage-by-Stage Log

Sessions covered: April 22, 2026 through May 13, 2026. Thirty-seven discrete stages documented.

Stage 1: Initial Ideation (April 22, 2026)

Topic	Evaluating class project concepts submitted via ChatGPT
Tool	Claude (Nina)
Claude Input	Received two AI startup pitch ideas from a teammate (Filippo): a regulatory copilot (RegRadar) and a B2B SaaS churn/next-best-action tool (PulseIQ). Asked Claude to challenge both concepts against the actual professor requirements: full-stack deployable prototype, token economics, GenAI Transparency Log, and a structural moat argument.
Claude Output	Identified that both moat arguments described features, not structural advantages. Built out two full pitch plans with pain point framing, token cost tables, business models, and prototype specs. Conducted competitive landscape research: Regology and Bolo AI as RegRadar competitors; Clari, Gainsight, MadKudu as PulseIQ competitors.
Human Contribution	Nina brought the original problem framing. The team decided which concerns to prioritise after seeing the critique. Strategic decision to focus on a defensible niche was made by the team, not by Claude.
Outcome	Neither idea was selected. The critique surfaced the core course requirement that moats must be structural — a principle that carried through every subsequent iteration.

Stage 2: Exploring Alternatives (April 23–26, 2026)

Topic	LinkedIn/job crawler, Career Navigator, energy compliance, grant validation
Tool	Claude (Nina)
Claude Input	The team tested several pivot concepts. For Career Navigator (Filippo's concept): asked Claude to evaluate a dual-mode career agent with persistent persona storage. For grant validation: asked whether an AI cross-checking tool existed in that space. For energy compliance: asked whether defensible niches remained open.
Claude Output	Career Navigator critique identified the memory feature as a feature-moat, not structural. Proposed the Contradiction Engine and Anti-Resume Persona Card as differentiators. Built a React prototype artifact for the Career Navigator demo. For grant validation: confirmed a gap exists in mid-process human handoff tools. For energy compliance: identified Regology and VComply as covering the enterprise segment, with mid-market niches open.

Human Contribution	Nina flagged the LinkedIn API legal risk independently. The team rejected all three concepts on feasibility and moat grounds after reviewing Claude's analysis — the final decision was theirs.
Outcome	None of these advanced. The repeated failure to find a structural moat pushed the team toward a more regulated, harder-to-replicate domain.

Stage 3: The Pivot – Construction RFI (April 27, 2026)

Topic	AI RFI Response Generator for construction projects
Tool	Claude (Nina)
Claude Input	A teammate (Paul) proposed an AI RFI generator for construction. The team asked Claude to challenge the concept and improve it.
Claude Output	The core RAG-over-documents logic was sound. Identified TrunkTools (\$70M Series B, US) and Civils.ai (UK) as direct competitors. Proposed the European construction market as an alternative beachhead, then identified Civils.ai as a UK competitor there too. Concluded the construction space was too crowded for a strong whitespace claim.
Human Contribution	Paul's original RFI concept. The team made the decision to keep the same RAG architecture but hunt for a domain where the whitespace claim was genuinely defensible.
Outcome	RFI for construction rejected. The RAG-over-regulatory-documents architecture was retained and applied to a new domain.

Stage 4: Domain Pivot – TSI Railway Compliance (April 27, 2026)

Topic	Identifying TrackMind as the core concept
Tool	Claude (Nina)
Claude Input	Following the construction dead end, Claude proposed EU railway TSI compliance as an alternative domain: same document intelligence architecture, zero direct competitors, structural moat from EU regulatory complexity.
Claude Output	Searched 403 AI railway startups (StartUs database). Confirmed zero competitors building TSI/NNTR compliance intelligence. Built the full TrackMind proposal document covering market research, pain point validation, unit economics, competitive landscape, moat analysis, tech stack, and 8-week MVP plan. Designed the IberRail S.A. demo scenario (Spanish manufacturer, French RFN cross-border authorisation) with two real documented conflicts: EN 14752 vs NF F31-054 obstacle detection protocol, and the CAS single-agent operation gap.
Human Contribution	The team evaluated and accepted the pivot. Nina reviewed all content for accuracy and business logic. Team validated the IberRail demo scenario. Edits were made to financial projections and customer targeting based on team judgment.

Outcome	TrackMind selected as the final project concept. The proposal document became the foundation for all subsequent work.
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Stage 5: Repository Setup and UI Tooling (April 28–29, 2026)

Topic	GitHub repo structure, license selection, .gitignore, UI framework choice
Tool	Claude (Nina)
Claude Input	Asked Claude to help set up the GitHub repository (ninaberenice/TrackMind_advML_TeamL). Questions on license selection, .gitignore configuration, and whether to use Figma vs. Lovable/Bolt for the frontend.
Claude Output	Recommended MIT license for an academic open project. Recommended Node + manual additions for .gitignore to cover Python, macOS, and secrets. Explained the distinction between Figma (design, not shippable) and vibe-coding tools (Lovable, Bolt — shippable React). Noted Streamlit as the simplest deployable option for the TrackMind prototype given the compliance workflow.
Human Contribution	Nina set up the repository. Team selected Streamlit as the frontend approach for the prototype. Team made the call to use VS Code as PyCharm fallback when branch sync issues arose.
Outcome	Repository live at ninaberenice/TrackMind_advML_TeamL. Dev environment confirmed: .venv, PyCharm primary, VS Code fallback, FastAPI via uvicorn port 8000.

Stage 6: Presentation Structure (April 30, 2026)

Topic	Structuring the 10-minute pitch for the course format
Tool	Claude (Nina)
Claude Input	Asked Claude to structure the 10-minute presentation balancing pitch logic, technical depth, live demo, and backend glimpse — matching the professor's dual requirement of seeing both the work and the business case.
Claude Output	Produced an 8-segment timing structure: Hook (1:30), Business Case (2:00), Architecture (1:30), Live Demo + Code (3:00), Moat + Close (2:00). Identified that the AI judge would specifically probe wrapper-startup arguments and prepared answers for the five most likely hard questions.
Human Contribution	Nina confirmed the team's comfort with the demo flow. The team assigned presenter names to each segment. The live demo scope (IberRail query, three-panel retrieval, CONFLICT DETECTED response) was confirmed by Nina.
Outcome	Presentation structure locked. Timing budget set at 10:00 total.

Stage 7: Project Alignment and Scope Clarification (May 3, 2026)

Topic	Understanding the team discussion and clarifying TrackMind's demo logic
Tool	ChatGPT (Paul)
ChatGPT Input	Paul shared a missed team discussion about whether the project should focus on manufacturers or compliance bodies, whether the manufacturer technical specification should be pre-ingested, and whether the app should present compliance functionality as the main product or as a future expansion. He asked ChatGPT to explain what he had missed and clarify the product logic.
ChatGPT Output	Summarised the discussion and clarified the key architectural issue: TrackMind should not treat the manufacturer SpecDoc as a permanently pre-ingested document if the product is intended to serve different manufacturers and different certification projects. Clarified that the TSI and French national rules could reasonably be stable pre-ingested sources because they are public regulatory documents. By contrast, the manufacturer specification is project-specific and should be uploaded or attached to a particular assessment session. Framed TrackMind as a compliance intelligence layer for cross-border railway authorisation rather than a static three-document demo.
Human Contribution	Paul identified the ambiguity in the team discussion and pushed for clarification on whether the system was mainly for manufacturers, certification bodies, or both. He also questioned the assumption that all three document types should always be pre-ingested. This was important because it exposed a real product-design issue: a manufacturer's technical specification varies by client and project, while the TSI and national rules are more stable regulatory sources.
Outcome	The product logic became clearer: TrackMind should work with two baseline legal sources (TSI and NNTR) and later incorporate manufacturer-specific technical specifications when uploaded. This directly shaped Paul's backend task and safety framing.

Stage 8: Understanding the Existing App Logic (May 6, 2026)

Topic	SpecDoc, bge-m3, ChromaDB, and current product behaviour
Tool	ChatGPT (Paul)
ChatGPT Input	Paul explained that he was looking at the code and project folder for the first time and did not fully understand the notebooks, backend logic, or how the components interacted. He asked ChatGPT to explain what a SpecDoc is, what bge-m3 is, what ChromaDB is, and how the app should behave if only two documents are available before a manufacturer uploads its own specification.
ChatGPT Output	Explained core technical concepts: SpecDoc is the manufacturer's technical specification, not a general regulation. bge-m3 is the multilingual embedding model used to convert English and French text into comparable vector representations.

	ChromaDB is the vector database storing chunks of regulatory and technical documents. The app must distinguish between legal-source reasoning and project-specific compliance assessment. With only TSI + NNTR, the app can answer general regulatory questions. Without an uploaded manufacturer specification, the app must not claim that a specific train design is compliant or non-compliant. Also identified the main risk: the system could accidentally imply it had manufacturer-specific evidence even when no SpecDoc had been uploaded.
Human Contribution	Paul recognised the central product risk: the demo still needed to work without the third file, but must not hallucinate that the third file exists. He pushed for a distinction between a general regulatory answer and a project-specific assessment. This was a human judgment call — it connected backend behaviour to trust, safety, and demo credibility.
Outcome	A clearer two-mode logic emerged: General regulatory mode: TSI + NNTR only. The app can explain applicable requirements and possible gaps, but cannot assess a specific train design. Project-specific assessment mode: TSI + NNTR + uploaded manufacturer SpecDoc. The app can compare the manufacturer's design or test evidence against the relevant rules. This distinction became a core part of Paul's contribution to the backend and safety logic.

Stage 9: Backend Architecture and Optional Spec Upload (May 6, 2026)

Topic	/upload-spec, /analyse, session cache, and avoiding persistent manufacturer data
Tool	ChatGPT (Paul)
ChatGPT Input	Paul shared backend code and comments describing an updated FastAPI architecture with /upload-spec, /analyse, /decide, and audit routes. The code stated that the spec PDF should be chunked in memory, stored in a session cache, and not written to ChromaDB. Paul asked ChatGPT to explain the architecture and what should change according to his task.
ChatGPT Output	Explained why the updated architecture mattered: /upload-spec should accept the manufacturer specification as a temporary session-level document. /analyse should use session-specific spec chunks only if a spec has actually been uploaded. The manufacturer specification should not be written to the persistent vector database by default. TSI and NNTR can remain persistent because they are public regulatory sources.

	Manufacturer specs may contain sensitive client or engineering information and should be handled as temporary project data. The UI should clearly show whether the current analysis is based on two sources or three sources. Clarified expected app behaviour: if no spec is uploaded, the system answers only from regulatory sources and explicitly states that no manufacturer specification is available. If a spec is uploaded, the system can compare manufacturer design claims against TSI and NNTR requirements.
Human Contribution	Paul was responsible for understanding and shaping the backend task. He forced the practical distinction between 'the demo should still work' and 'the demo must not pretend a spec exists.' That distinction was strategically important because it prevented TrackMind from appearing hallucination-prone and strengthened the GDPR data-minimisation story by treating manufacturer specs as session-specific rather than permanently stored.
Outcome	The proposed backend logic became: persistent legal collections for TSI and NNTR, plus temporary session-based manufacturer spec chunks. This improved privacy, trustworthiness, and hallucination control.

Stage 10: Technical Build – Embeddings, Ingestion, Retrieval (May 7, 2026)

Topic	Full RAG pipeline implementation: BAAI/bge-m3, ChromaDB, ingest.py, retrieval.py
Tool	Claude (Nina)
Claude Input	Asked Claude for end-to-end technical guidance on implementing the tri-source RAG pipeline. Key decisions to make: embedding model selection (OpenAI vs. open-source), national rules document sourcing, ChromaDB collection structure, IberRail SpecDoc content, and cross-lingual retrieval logic.
Claude Output	Recommended BAAI/bge-m3 (HuggingFace, free, multilingual, 1024 dimensions) over OpenAI embeddings to support cross-lingual retrieval between French NNTRs and Spanish technical specs. Identified the correct TSI document (CELEX_02014R1302-20250427_EN_TXT.pdf). Found that EUR-Lex CELEX URLs return 202/HTML rather than PDFs and Légifrance returns 403 — both require manual download. Identified the French Arrêté du 19 mars 2012 (Article 49) as the best real national rules document. Wrote ingest.py (article-level chunking with fixed-size fallback, bge-m3 embeddings, three ChromaDB collections: tsi_loc_pas, nntr_france, spec_doc). Wrote retrieval.py (tri-source retrieval with context builder, cross-lingual test queries). Generated four versions of SpecDoc.pdf for IberRail Rev. 4.0 with both documented conflicts. Updated the TrackMind Proposal document to reflect all technical changes.
Human Contribution	Nina manually downloaded all three source documents and placed them in docs/. Confirmed bge-m3 model loaded successfully (1024 dimensions). Confirmed ingestion was running. Nina corrected Claude's terminology twice: once noting 'regulation' alone undersells the product by omitting TSIs; once proposing her own tagline variant that Claude

Outcome	built on. Nina flagged that document outputs should not be shorter than prior versions.
	ingest.py and retrieval.py complete. Three ChromaDB collections populated. Tech stack confirmed: bge-m3 + ChromaDB + Claude via Anthropic API key. Virtual environment at .venv.

Stage 11: API Key and Model Selection (May 8, 2026)

Topic	Anthropic API key confirmation; reasoning layer model choice
Tool	Claude (Nina)
Claude Input	Nina asked whether an Anthropic API key (sk-ant- prefix) would work for the project, and how to identify which provider the key belonged to.
Claude Output	Confirmed the key was Anthropic (sk-ant- prefix, console.anthropic.com). Recommended swapping GPT-4o-mini for claude-haiku-4-5-20251001 as the reasoning layer: faster, cheaper, and well-suited for structured regulatory reasoning with a well-engineered system prompt. Provided the Python API call pattern. Noted that unit economics cost-per-query would shift slightly (Haiku is cheaper than GPT-4o-mini).
Human Contribution	Nina identified the API key from the Anthropic Console screenshot. Team confirmed the switch to Claude as the reasoning layer. Unit economics were updated accordingly.
Outcome	Reasoning layer confirmed as Claude (claude-haiku-4-5-20251001 for demo, claude-sonnet-4-6 for benchmark evaluation).

Stage 12: Technical Risk Review and Demo Reliability (May 8, 2026)

Topic	Mock mode, persistent SpecDoc risk, audit traceability, and session lifecycle
Tool	ChatGPT (Paul)
ChatGPT Input	Paul asked ChatGPT to review technical risks in the current TrackMind prototype. The key concern was that the app might generate misleading compliance conclusions if no manufacturer specification had been uploaded. Specific issues included mock-mode behaviour, possible old spec_doc collections in ChromaDB, missing audit metadata, and unclear handling of temporary uploaded specification files.
ChatGPT Output	Identified the most important technical risks, structured into priority areas. Must fix before demo: Mock-mode risk: mock responses should not return spec-specific conclusions when no manufacturer SpecDoc is available. Otherwise, the demo could falsely imply that TrackMind has reviewed manufacturer evidence. Persistent-spec risk: old ingestion workflows may have written a spec_doc collection into ChromaDB. This should be checked and removed or clearly separated from the live session upload logic.

	Audit traceability: audit logs should include metadata to reconstruct which document context was used, including whether a session-specific spec was present. Should fix if time: Add timestamps or expiry logic for temporary in-memory spec files. Make evidence labels clearer in the UI. Improve fallback behaviour when the system has legal sources but no manufacturer-specific technical evidence.
Human Contribution	Paul identified the key credibility risk: the system must still be useful with only TSI and NNTR documents, but must not claim manufacturer-specific compliance without an uploaded SpecDoc. He pushed the distinction between general regulatory reasoning and project-specific compliance assessment — central to making the demo honest and defensible.
Outcome	The technical task became more precise. TrackMind should behave differently depending on whether a manufacturer specification is available. This improved the safety logic, privacy story, and demo credibility.

Stage 13: Logo and Visual Assets (May 8, 2026)

Topic	Removing white background from TrackMind logo for slide use
Tool	Claude (Nina)
Claude Input	Uploaded the TrackMind logo PNG and asked Claude to remove the white background and produce a transparent version for slide use.
Claude Output	Used Python Pillow to mask pixels above RGB (230, 230, 230) and set alpha to 0. Output: TrackMind_logo_transparent.png.
Human Contribution	Nina uploaded the original logo. The logo design itself was created by the team independently.
Outcome	Transparent logo PNG produced and used in the pitch deck.

Stage 14: Pitch Deck – First Draft (May 9, 2026)

Topic	Building TrackMind_AI_Presentation.pptx from the presentation structure document
Tool	Claude (Nina)
Claude Input	Uploaded the Presentation_Structure.docx and a TrackMind template PPTX. Asked Claude to build the first-draft presentation using the template's colour palette and mapped to the 8-segment timing structure.
Claude Output	Read the template thumbnail (navy #0C2336, teal #387C9D, sage green #729D72). Produced a 10-slide PPTX covering: Title/Hook, Problem in Numbers, Business Model + Token Economics, Architecture Diagram, Live Demo (IberRail scenario with three retrieved chunks and CONFLICT DETECTED response), Benchmark Methodology, Moat Table, Safety

Human Contribution	Architecture (GREEN/AMBER/RED confidence tiers), Competitive Landscape, GenAI Transparency + Close.
	The Presentation_Structure.docx with detailed segment plans was created by the team. Team members were assigned to specific slides. Nina flagged placeholder fields (team names, live benchmark numbers) to fill in before submission.
Outcome	TrackMind_AI_Presentation.pptx first draft complete.

Stage 15: Business Case Document – Competitive and Market Research (May 9–10, 2026)

Topic	Market scale, competitive landscape, AI platform threat sections
Tool	Claude (Nina)
Claude Input	Asked Claude to consolidate two business case documents (TrackMind_Business_Model.docx and TrackMind_BusinessCase_Draft.docx) and add new sections: Section 1.4 (Market Scale), Section 2.4 (Competitive Landscape), Section 2.5 (AI Platform Threat Analysis).
Claude Output	Conducted web searches: ERA has authorised ~80,000 vehicles since 2019; European rolling stock market ~€120 billion; 40+ NoBos operate across Europe. Section 2.5 analysed Anthropic, OpenAI, Harvey AI, and frontier model substitution risk. Produced TrackMind_Complete_EN_v2.docx then TrackMind_Complete_EN_v3.docx. Also rebuilt the pitch presentation as TrackMind_Presentation_v2.pptx with 11 slides, presenter names in each footer, and timing mapped to the 8 segments.
Human Contribution	Nina confirmed the go-to-market sequencing: Phase 1 targets mid-size rolling stock manufacturers; Phase 2 expands to NoBos. This was a deliberate strategic decision reconfirmed by Nina and treated as settled. Nina reviewed all financial projections and market sizing against primary sources.
Outcome	TrackMind_Complete_EN_v3.docx and TrackMind_Presentation_v2.pptx are the current live versions.

Stage 16: Tagline Exploration (May 10, 2026)

Topic	Website hero copy and pitch tagline options
Tool	Claude (Nina)
Claude Input	Asked Claude to generate tagline options for the TrackMind website hero. Nina proposed her own variant mid-session: 'We Read the Regulation and Find the Conflict. You Ship the Train.'
Claude Output	Generated multiple rounds of tagline options across tone categories (direct/utility, dramatic, technical credibility, minimal, question-led). Produced a consolidated list organised by category. Built on Nina's

Human Contribution	proposed variant as the strongest direction, noting the rhythm of three distinct actions and the precision of distinguishing TSIs from NNTRs.
	Nina proposed her own tagline variant, corrected Claude on accuracy (noting that 'regulation' alone omits TSIs), and selected the final approach direction. The tagline selection itself was made by the team.
Outcome	Tagline options list produced. Final selection to be confirmed by the team before submission.

Stage 17: Technical – Branch Navigation and VS Code (May 10–11, 2026)

Topic	Checking out teammate branches without modifying their work
Tool	Claude (Nina)
Claude Input	Nina needed to inspect Catarina's UI/UX branch and Afonso's testUpdates branch on the shared repo. Asked for step-by-step guidance tied to what was visible on screen.
Claude Output	Guided through: git fetch origin, git checkout for each branch, read-only inspection. Identified that PyCharm was not reflecting remote branches after fetch. Recommended VS Code as the fallback, where the branch appeared in the bottom-left status bar.
Human Contribution	Nina navigated the terminal and IDE. The branch checkout decisions were hers. Nina confirmed VS Code resolved the PyCharm display issue.
Outcome	VS Code confirmed as the reliable fallback for Git branch operations. FastAPI server confirmed running via uvicorn at port 8000.

Stage 18: Pitch Deck and Presentation Refinement (May 10–12, 2026)

Topic	Safety slide, live demo slide, GTM wording, and value-add framing
Tool	ChatGPT (Paul)
ChatGPT Input	Paul used ChatGPT to refine several presentation elements for the TrackMind pitch deck. He asked for content for the safety and regulatory compliance slide, including AI Act and GDPR boxes plus safety pillars. He also asked for live demo slide bullets, demo titles, and shorter action titles for the Go-to-Market slide. Throughout the process, Paul corrected weak wording and pushed for more direct, slide-ready language.
ChatGPT Output	For the Safety and Regulatory Compliance slide: AI Act section: Human-in-the-loop (TrackMind drafts, the assessor approves); high-risk awareness (railway compliance requires controlled AI support); auditability (decisions must be traceable to retrieved evidence and reviewer action). GDPR section: Data minimisation (only process documents needed for the compliance query); session-based spec handling (manufacturer

	specs should not be stored permanently by default); traceability without over-collection (logs preserve decisions, not unnecessary personal data). Safety pillars: Grounding (answers based on retrieved TSI, NNTR, and uploaded spec evidence); Escalation (low-confidence or missing-evidence cases routed to human review); Traceability (queries, retrieved chunks, confidence, and reviewer decisions are logged); Separation (public legal sources stay persistent; manufacturer specs remain session-specific). For the Live Demo slide: shifted wording from technical walkthrough toward user value workflow — upload, retrieve, detect conflicts, escalate safely, approve with audit trail. For the Go-to-Market slide: shorter action-title alternatives — Land with Manufacturers; Start with Pre-Assessment; Scale to NoBos.
Human Contribution	Paul strongly shaped this stage. He rejected generic slide wording, corrected the structure multiple times, and pushed for content that matched the actual product logic. He insisted that safety should not be presented as abstract regulation talk — it needed to be shown as concrete product behaviour: retrieval grounding, optional SpecDoc upload, no hallucinated manufacturer evidence, confidence gating, and human approval. Paul also redirected the live demo framing toward value creation rather than functionality description.
Outcome	The pitch deck became more coherent and credible. The safety slide connected directly to TrackMind's backend design, while the live demo slide became easier for listeners to understand. The presentation moved from 'this is how the tool works' toward 'this is why the tool creates value and reduces risk.'

Stage 19: Business Analysis, Economic Modelling, and Strategic Document (May 9–12, 2026)

Topic	Full business case document: market context, player analysis, unit economics, revenue model, financial projections, go-to-market roadmap, data protection, regulatory corpus strategy, EU AI Act compliance
Tool	Claude (Filippo)
Claude Input	Filippo provided the original LLM-generated business proposal as a starting point, then directed Claude through a structured multi-session analysis covering: market context and regulatory framework; player ecosystem (manufacturers, NoBos, ERA); customer selection rationale; cost equation for the target customer; unit economics in two versions (academic prototype and hypothetical production product); revenue model design; financial projections for Year 1–3; go-to-market roadmap; data protection architecture; regulatory corpus maintenance strategy; EU AI Act risk classification; and GDPR compliance posture. Claude also applied iterative revisions based on Filippo's review comments — including a full document uploaded with yellow-highlighted corrections and Italian-language notes — and merged multiple document versions across the project lifecycle.
Claude Output	Produced a complete business analysis document (approximately 800 paragraphs, fully in English) covering: section-by-section analysis of the

	cross-border railway certification market and its three players; justified selection of mid-size manufacturers as the Phase 1 beachhead; cost equation for the target customer with verified benchmarks (ERA hourly rate EUR 265/hour, engineer salary, NoBo billing structure); dual unit economics analysis separating the academic prototype (EUR 0.033/query, all other costs zero) from the hypothetical production product (EUR 14.50/month silicon + EUR 1,083/month carbon per client at 5 clients); revenue model with pricing justified from customer ROI (EUR 1,000/month manufacturers, EUR 2,500/month NoBos); three-year financial projections with explicit Year 1 loss justification; three-phase go-to-market roadmap with large manufacturers removed beyond the 3-year horizon; a data protection framework positioning ChromaDB self-hosted as a competitive differentiator over managed vector database alternatives; a three-layer regulatory corpus maintenance strategy with a contractual SLA on corpus currency; EU AI Act Limited Risk classification with full argumentation; and a GDPR compliance framework. Infrastructure and salary benchmarks were verified against primary sources (Anthropic official pricing, GCP pricing, Portuguese employer contribution law, Boundless Employment Costs Report 2025).
Human Contribution	Filippo's contribution was substantive throughout. He identified the weaknesses in the original LLM-generated document and insisted on separating the academic prototype economics from the hypothetical product — a distinction that materially changed the analytical structure. All key strategic decisions were his: the choice of ChromaDB self-hosted over Weaviate Cloud (and its data privacy implications), the decision to price at EUR 1,000/month flat rather than per-query, the removal of large manufacturers from the roadmap to reduce the attack surface on the business plan, and the acceptance of a planned Year 1 loss rather than artificially raising the price to break even. Filippo performed fact-checking throughout, pushed back on assumptions he found unrealistic, simplified overly complex calculations, and conducted two full review sessions identifying numerical inconsistencies and narrative gaps. The framing of data protection and regulatory currency as competitive differentiators — not merely compliance obligations — was directed by Filippo.
Outcome	Complete business analysis document used as the analytical foundation for the TrackMind business case, incorporated into the team's final submission materials.

Stage 20: GenAI Transparency Log (May 12–13, 2026)

Topic	This document
Tool	Claude (Nina)
Claude Input	Asked Claude to compile the complete GenAI Transparency Log covering all sessions from initial ideation through the current state, and to compare Claude's contributions against Nina's own inputs. Paul's ChatGPT session data was subsequently integrated to produce the final merged log.
Claude Output	Retrieved conversation history across 14 sessions (April 22 – May 12, 2026). Produced a structured log document covering 14 discrete stages for Nina, 1 for Filippo, and 5 for Paul, with input/output analysis per stage, a human vs. AI contribution comparison, and a summary of key

Human Contribution	principles that emerged from the collaboration. Paul's five ChatGPT stages and Filippo's one Claude stage were integrated chronologically to produce this final merged document.
	Nina provided the project brief, all domain corrections, strategic direction, document review, and the instruction to compile this log. Paul defined the scope of his ChatGPT usage, corrected the structure, and identified which sessions were relevant to include. The framing of what counted as 'human contribution' vs. 'AI output' was the judgment call of both contributors throughout.
Outcome	This document — a single integrated GenAI Transparency Log for the full TrackMind team, covering both Claude (Nina) and ChatGPT (Paul) usage.

Stage 21: UI Prototyping – Initial Interface and Frontend Refinements (May 2026)

Topic	TrackMind web interface: initial HTML generation and frontend refinements to demo.html and index.html
Tool	Lovable (AI website builder) + Claude (Catarina)
Claude/AI Input	Lovable generated first-draft HTML for demo.html and index.html from Catarina's product description. Claude was then used iteratively to refine demo.html: PDF export spacing, confidence level badge redesign, layout of Steps 1–3, block widths, analysis time position. Also organised index.html and styles.css and fixed hero animation.
Claude/AI Output	Lovable produced a rough structural starting point for both HTML files. Claude refined layout and styling across multiple iterations, reorganised CSS for readability, and removed unused CSS rules without altering visual output.
Human Contribution	Catarina provided the product description to Lovable and directed every design change through specific instructions. She decided to extract styles into a separate styles.css file, tested all results in-browser, and validated no visual regressions before accepting each output.
Outcome	demo.html, index.html, and styles.css finalised. Confidence badge, Step layout, PDF spacing, and hero animation working correctly.

Stage 22: Backend Logic – Document Context Alignment and Cross-Source Retrieval (May 2026)

Topic	Aligning Document Context panel with verdict output; cross-reference expansion for regulatory documents
Tool	Claude (Catarina)
Claude Input	Two problems: (1) Document Context panel was selecting chunks before the LLM responded, causing misalignment with cited sources. (2) When a SPEC chunk mentioned NF F31-054 or EN 14752, the corresponding regulatory document was not being pulled into Citations.

Claude Output	(1) Redesigned api.py and demo.html so chunks are selected after the LLM responds. (2) Implemented cross-source reference expansion logic in the retrieval pipeline.
Human Contribution	Catarina designed both architectural changes and directed implementation. She specified exact behaviour and edge cases for cross-reference expansion, and validated results against benchmark questions.
Outcome	Document Context shows only chunks cited in the model's explanation. Cross-source references (NF F31-054, EN 14752) correctly trigger retrieval of corresponding regulatory documents.

Stage 23: Backend Logic – Retrieval Fixes and Confidence Calibration (May 2026)

Topic	Diagnosing retrieval misses for TSI 4.2.5.5.9 and correcting wrong AMBER confidence on TSI-only queries
Tool	Codex / GPT-4.1 via VS Code (Catarina)
AI Input	Two issues: (1) LOC&PAS TSI section 4.2.5.5.9 not retrieved on relevant queries. (2) Confidence calibration applying an incorrect French RFN penalty to TSI-only queries.
AI Output	Codex added a priority boost for article numbers explicitly mentioned in the query (retrieval.py). Removed the French RFN penalty from confidence calibration in reasoning.py for TSI-only queries.
Human Contribution	Catarina identified both problems from test runs and specified the exact fix for each. She validated results after each change.
Outcome	TSI 4.2.5.5.9 retrievable. TSI-only queries no longer penalised. Confidence calibration corrected.

Stage 24: Code Quality – Dead Code Removal and Response Evaluation (May 2026)

Topic	Removing dead code from reasoning.py; evaluating compliance question responses
Tool	Claude (Catarina)
Claude Input	reasoning.py contained unused private helper functions. Separately, Claude analysed compliance question responses to identify misaligned citations, incorrect confidence levels, and unsupported references.
Claude Output	Performed AST analysis identifying four unused functions and removed them. Produced assessments of compliance response quality with specific issues flagged.
Human Contribution	Catarina requested the cleanup, reviewed the AST analysis to confirm no functional regressions, used the evaluations to direct further fixes, and made all accept/reject decisions.

Outcome	reasoning.py cleaned of dead code. Compliance response quality issues identified and addressed.
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Stage 25: Documentation – Catarina’s GenAI Transparency Log (May 2026)

Topic	Drafting Catarina’s GenAI Transparency Log
Tool	Claude (Catarina)
Claude Input	Asked Claude to draft the transparency log based on conversation history and Codex logs provided by Catarina.
Claude Output	Produced a structured log covering all AI-assisted development phases, activity table, reflections, and academic integrity statement.
Human Contribution	Catarina provided all source logs, corrected framing throughout, specified structure, and reviewed the final output.
Outcome	GenAI Transparency Log for Catarina’s contribution completed and integrated into this merged document.

Stage 26: API Switch – Gemini to Claude Sonnet (May 13, 2026)

Topic	Replacing Gemini API with Anthropic Claude Sonnet in the reasoning layer
Tool	Claude (Afonso)
Claude Input	Gemini was returning persistent 429 rate limit errors. Asked Claude to rewrite <code>_call_gemini()</code> to use the Anthropic REST API and diagnose why <code>max_tokens:1000</code> was causing truncation.
Claude Output	Rewrote <code>_call_gemini()</code> to <code>_call_claude()</code> using Anthropic <code>/v1/messages</code> . Identified <code>max_tokens:1000</code> as insufficient; recommended 3000. Identified hardcoded API key risk and provided <code>os.environ</code> pattern.
Human Contribution	Afonso identified the Gemini failure, made the provider switch decision, confirmed model string (<code>claude-sonnet-4-5</code>), ran the app to verify CONFLICT DETECTED responses, and updated API key handling.
Outcome	reasoning.py using Claude Sonnet via Anthropic API. <code>max_tokens</code> set to 3000. Hardcoded API key removed.

Stage 27: Session-Based Spec Upload Pipeline (May 13, 2026)

Topic	In-memory session-based spec upload so manufacturer PDFs never persist to ChromaDB
Tool	Claude (Afonso)
Claude Input	Asked Claude to design and implement a session-based spec upload system where manufacturer PDFs are never written to ChromaDB.

Claude Output	Designed <code>_SESSION_SPECS</code> dict keyed by UUID session IDs. Implemented <code>/upload-spec</code> POST, <code>chunk_generic_from_bytes()</code> , <code>DELETE /upload-spec/{session_id}</code> , and updated <code>/analyse</code> to resolve session spec chunks.
Human Contribution	The architectural decision that specs must never persist to ChromaDB was Afonso's — a deliberate privacy and product design choice. He reviewed all endpoint logic and confirmed the session ID flow end-to-end.
Outcome	<code>/upload-spec</code> endpoint live. <code>_SESSION_SPECS</code> in-memory store working. Spec never touches ChromaDB.

Stage 28: ChromaDB spec_doc Collection Removal (May 13, 2026)

Topic	Deleting stale <code>spec_doc</code> collection from ChromaDB left by earlier notebook ingestion
Tool	Claude (Afonso)
Claude Input	Teammate flagged risk that old ingestion had written a <code>spec_doc</code> collection to ChromaDB. Asked Claude to verify and delete it.
Claude Output	Provided one-liner to check collections, confirmed <code>spec_doc</code> present, provided delete command.
Human Contribution	Risk was identified by a teammate in code review. Afonso ran both commands and confirmed final state: only <code>tsi_loc_pas</code> and <code>nntr_france</code> remain.
Outcome	<code>spec_doc</code> collection permanently deleted. ChromaDB contains only <code>tsi_loc_pas</code> and <code>nntr_france</code> .

Stage 29: Section-Aware Spec Chunker and Header Stripping (May 13, 2026)

Topic	Replacing fixed 1500-char sliding window chunker with section-aware chunker for structured technical specifications
Tool	Claude (Afonso)
Claude Input	Existing <code>chunk_generic()</code> produced low-quality chunks for structured specs. Asked Claude to implement section-aware chunking with header stripping.
Claude Output	Implemented <code>_chunk_spec_text()</code> with regex splitting on numbered headers, OI-N open issues, and V-N verification items. Added header noise filtering. Dynamic <code>source_name</code> instead of hardcoded IberRail.
Human Contribution	Afonso identified the quality problem and specified the splitting strategy. The dynamic <code>source_name</code> decision was his. He tested with <code>SpecDoc.pdf</code> and confirmed 30 section-level chunks.
Outcome	<code>trackmind_chunker.py</code> updated. 30 meaningful section-level chunks produced. Header noise eliminated.

Stage 30: Pre-Computing Embeddings at Upload Time (May 13, 2026)

Topic	Moving bge-m3 embedding computation from query time to upload time to reduce first-query latency from ~77s to ~5–10s
Tool	Claude (Afonso)
Claude Input	First query after upload was taking 77 seconds. Asked Claude to move embedding to upload time.
Claude Output	Added embedding pre-computation loop in /upload-spec. Confirmed cache hit logic in retrieve_with_session_spec() would reuse values on subsequent queries.
Human Contribution	Afonso identified the 77s delay and made the trade-off decision (slower upload for faster queries). He confirmed the cache hit logic was correct.
Outcome	Upload takes ~40s. First query reduced from ~77s to ~5–10s. Subsequent queries use cached embeddings.

Stage 31: Full-Text Chunk Modal (May 13, 2026)

Topic	Adding modal to display complete chunk text, previously truncated to 300 characters in the Document Context panel
Tool	Claude (Afonso)
Claude Input	Document context chunks truncated to 300 characters. Asked Claude to implement a modal showing complete chunk text.
Claude Output	Implemented openChunkModal(), truncateClean(), View full text button per chunk, #chunk-modal HTML, and .btn-view-full CSS.
Human Contribution	Afonso specified the requirement, reviewed the implementation, identified and fixed a duplicate modal HTML bug, and confirmed the modal worked for both EN and FR chunks.
Outcome	Full-text modal working. Every chunk has a View full text button showing language flag, source reference, and complete chunk text.

Stage 32: Dashboard Cleanup – Removing Orphaned Escalated Stat (May 13, 2026)

Topic	Fixing JavaScript null reference error caused by orphaned Escalated counter after Escalate button removal
Tool	Claude (Afonso)
Claude Input	Removing the Escalate button left the Escalated counter orphaned, causing a JavaScript null reference error. Asked Claude to identify and fix the root cause.
Claude Output	Identified root cause: loadAuditStats() still referenced stat-escalated which no longer existed. Removed references and updated audit-stats grid from repeat(5,1fr) to repeat(4,1fr).

Human Contribution	Afonso identified the JavaScript error from the browser toast, confirmed the fix worked, and validated the 4-stat layout was correct.
Outcome	JavaScript error eliminated. Dashboard shows 4 stats: Accepted, Rejected, Pending, Estimated Time Saved.

Stage 33: Markdown Bold Stripping in Explanation Display (May 13, 2026)

Topic	Removing raw bold markdown markers appearing visually in the Explanation and Recommended Action panels
Tool	Claude (Afonso)
Claude Input	Claude Sonnet was returning bold markdown formatting causing asterisks to appear in the UI.
Claude Output	Updated formatNumberedHtml() to strip text markers with regex before rendering.
Human Contribution	Afonso identified the visual bug from a screenshot and verified the fix in the browser.
Outcome	Markdown bold markers no longer visible in UI output panels.

Stage 34: Verdict Parser Debugging (May 13, 2026)

Topic	Diagnosing intermittent INSUFFICIENT DATA verdicts when explanation clearly described CONFLICT DETECTED
Tool	Claude (Afonso)
Claude Input	Verdict intermittently showing INSUFFICIENT DATA. Added debug print to capture raw Claude response.
Claude Output	Suggested parser regex failure or max_tokens truncation as possible causes. Provided debug print pattern.
Human Contribution	Afonso added the debug print, ran the app, and identified that restarting uvicorn fixed the issue — stale server state was the actual root cause. Claude's diagnosis was incorrect; Afonso's operational knowledge led to the correct fix.
Outcome	Server restart confirmed as fix. Verdict parsing working correctly.

Stage 35: Mock Mode Removal (May 13, 2026)

Topic	Hiding mock mode button and setting live API as default to prevent misleading demo behaviour
Tool	Claude (Afonso)
Claude Input	Mock mode button visible and could return hardcoded SPEC-specific answers without a spec uploaded, making the demo appear dishonest. Asked Claude to hide the button and fix initMode().

Claude Output	Added display:none to mock mode button HTML. Updated initMode() to hardcode live instead of reading localStorage.
Human Contribution	Afonso made the product decision to remove mock mode. The credibility rationale was his judgment, confirmed by teammate code review.
Outcome	Mock mode hidden. initMode() always starts in live mode.

Stage 36: Benchmark Evaluation and Results Interpretation (May 13, 2026)

Topic	Running benchmark and interpreting results in light of session-based spec privacy architecture
Tool	Claude (Afonso)
Claude Input	Asked Claude to advise on running the benchmark, identify compatibility issues, and help interpret results.
Claude Output	Confirmed benchmark.py compatible with current architecture. Explained that metrics 3 and 4 require a spec upload which the automated benchmark cannot provide, consistent with the privacy model.
Human Contribution	Afonso ran benchmark.py --retrieval-only and obtained results (TSI 89%, NNTR 86%, Cross-lingual 80%). He decided to frame the metrics 3 and 4 limitation as a privacy architecture feature rather than an evaluation gap.
Outcome	Benchmark results: TSI 89% PASS, NNTR 86% PASS, Cross-lingual 80% PASS. Full benchmark: 5/5 metrics passing. benchmark_results.json committed.

Stage 37: README and Architecture Slides (May 13, 2026)

Topic	Generating repository README and interactive architecture slide visuals
Tool	Claude (Afonso)
Claude Input	Asked Claude to generate a comprehensive README.md and interactive architecture visuals covering the full pipeline, cross-lingual retrieval mechanism, and key design decisions.
Claude Output	Generated README.md covering architecture, setup, demo flow, API endpoints, benchmark results, limitations, and tech stack. Generated interactive SVG pipeline diagram and cross-lingual retrieval diagram.
Human Contribution	Afonso reviewed and confirmed all technical details. He provided the content for the architecture slide pipeline table and made all decisions about what to include in the presentation vs written report.
Outcome	README.md complete and committed. Architecture slide content finalised.

Human Contribution vs. AI Output: Comparison

The table below assesses each major work area against two questions: what did the AI tool produce, and how much did human judgment shape or override that output? The 'Human Weight' column is the team's honest assessment.

Work Area	What AI Produced	Human Weight in Final Output
Startup concept generation (Claude)	Proposed TrackMind pivot after the RFI construction domain was rejected as too crowded	Medium — the domain idea came from Claude, but the team evaluated and accepted it
Competitive landscape research (Claude)	Searched 403 railway startups and confirmed zero direct competitors	Low — Nina verified key claims against primary sources
Financial model structure (Claude)	Designed unit economics tables, revenue model, and projection structure	Medium — team reviewed projections and edited customer targeting sections
Go-to-market sequencing (Claude)	Proposed initial GTM structure; team reversed Phase 1 and Phase 2 order	High — Nina's decision to lead with manufacturers before NoBos was a strategic override
Technical architecture (Claude)	Recommended bge-m3, ChromaDB collections, article-level chunking, retrieval.py structure	Medium — Nina implemented the code, downloaded documents manually, confirmed model loaded
Document sourcing (Claude)	Identified the correct TSI document and the French Arrêté du 19 mars 2012	High — EUR-Lex and Légifrance both require manual download; Nina sourced all three files
SpecDoc content (Claude)	Generated four versions of the IberRail SpecDoc.pdf with both documented conflicts	Medium — Nina confirmed the IberRail narrative and the French RFN scenario over a Spanish NNTR
Terminology accuracy (Claude)	Initial outputs used 'regulation' alone; Nina corrected to include 'TSIs and national rules'	High — Nina's domain corrections shaped all subsequent language
Tagline creation (Claude)	Generated tagline options across tone categories	High — Nina proposed her own variant mid-session that became the strongest candidate
Presentation structure (Claude)	Built the 8-segment timing structure and slide-by-slide content	Medium — Presentation_Structure.docx with segment plans was created by the team beforehand
Pitch deck PPTX (Claude)	Generated both PPTX versions using the template's colour palette	Medium — team designed the template, assigned presenter names, confirmed demo flow
Moat argument framing (Claude)	Structured the moat analysis (regulatory complexity, data flywheel, EU AI Act)	Medium — team reviewed and accepted the framing; no challenges or overrides on this point
Safety architecture design (Claude)	Designed the GREEN/AMBER/RED confidence gating and audit trail	Low — accepted as-is; directly reflects the course's human-in-the-loop requirement

Work Area	What AI Produced	Human Weight in Final Output
Project alignment (ChatGPT)	Summarised missed team discussion and clarified manufacturer vs. compliance-body logic	High — Paul identified the ambiguity and asked the right questions
SpecDoc upload logic (ChatGPT)	Explained why the manufacturer spec should be optional and session-based	High — Paul pushed the 'works with two docs but must not hallucinate the third' requirement
Backend architecture (ChatGPT)	Interpreted /upload-spec, /analyse, session cache, and persistent ChromaDB risks	Medium-high — ChatGPT explained the design; Paul's task ownership drove the final logic
Demo reliability (ChatGPT)	Structured risks around mock mode, persistent SpecDoc collections, audit metadata, and temporary spec handling	High — Paul identified the credibility risk and shaped the priority order
Safety slide (ChatGPT)	Generated AI Act, GDPR, and safety pillar content	High — Paul rejected poor structure and forced concise, slide-ready wording
Demo and GTM slide wording (ChatGPT)	Suggested live demo bullets, value-add framing, and shorter GTM action titles	Medium — ChatGPT gave wording options; Paul selected direction and corrected weak framing
Business case, unit economics, financial model (Claude)	Produced the full business analysis document: market context, player analysis, customer selection rationale, dual unit economics (prototype and production), revenue model with ROI-based pricing, three-year financial projections, go-to-market roadmap, data protection framework, regulatory corpus maintenance strategy, EU AI Act classification, GDPR posture. All figures verified against primary sources.	Medium-high — Claude produced the analytical content, but Filippo set the direction for every section, systematically challenged all proposals, and made all strategic decisions. The framing of key differentiators (data protection, regulatory currency) was directed by Filippo.

Key Principles That Emerged from the Collaboration

Several recurring themes shaped how the team worked with AI tools throughout the project. These are documented here because they reflect genuine intellectual contributions made during the process, not just outputs.

1. Moat arguments must be structural, not feature-based

This became the filter applied to every concept explored. The Career Navigator's persistent memory, the construction RFI's document search — both were features another product could add in a sprint. TrackMind's moat (EU regulatory complexity, 18-month rebuild time, data flywheel across 27 national rule sets) is structural. Claude proposed this framing early; the team internalised it and applied it independently to each subsequent pivot.

2. Domain precision matters more than general accuracy

Nina corrected Claude's language multiple times when it used 'regulation' in contexts where the full set is 'TSIs, NNTRs, and manufacturer technical specifications.' These corrections were not pedantic — each one sharpened the value proposition and tightened the competitive claim.

3. Real documents over fictional ones

The decision to use the actual LOC&PAS TSI (CELEX_02014R1302-20250427) and the real French Arrêté du 19 mars 2012 rather than synthetic documents was a deliberate quality decision made by the team. Claude identified the correct documents; Nina did the manual work of downloading them when automated retrieval failed.

4. The assessor signs everything

The human-in-the-loop architecture was non-negotiable from Stage 4 onward. Every confidence tier, every audit log entry, every demo flow ends with a licensed NoBo assessor approving the draft. This is both a safety design and a liability design — the two are inseparable in the TrackMind context.

5. The app must not pretend the manufacturer spec exists

Paul's most important architectural contribution. A weaker version of the project would demo three documents as if the SpecDoc were always available. Paul identified the more realistic product logic: TSI and NNTR can be stable baseline sources, but the manufacturer spec is client-specific and should be uploaded per session. This changed the backend logic, the UI expectations, and the safety story.

6. Safety is product behaviour, not slide content

The strongest safety argument came from connecting slide content to backend logic: no persistent manufacturer spec by default, no spec-based claims without upload, confidence gating, audit metadata, and human approval. This is substantially stronger than generic 'we follow GDPR and AI Act' language.

7. Mock mode can destroy demo credibility

A canned response that claims SpecDoc evidence without an actual spec upload makes the demo look dishonest — exactly the kind of detail a critical evaluator could penalise. A polished interface is not sufficient if the underlying behaviour is misleading.

8. The demo must communicate value, not just functionality

The final demo framing shows the full value chain: upload, retrieve, compare, detect conflict, escalate, approve, log. That is the product story. Without this framing, the demo risks appearing as a technical prototype rather than a business solution.
